Multimodal Disinformation Detection with HAMMER

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Prof Michael Brünig,
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Dear Professor, Brünig,

In accordance with the requirements of the Degree of Bachelor of Engineering (Honours) in the School of Electrical Engineering and Computer Science, I submit the following thesis entitled

"Multimodal Disinformation Detection with HAMMER"

The thesis was performed under the supervision of Dr. Priyanka Singh and Professor Xue Li. I declare that the work submitted in the thesis is my own, except as acknowledged in the text and footnotes, and that it has not previously been submitted for a degree at the University of Queensland or any other institution.

Yours sincerely

Gagandeep Singh

Acknowledgments

I would like to express my sincere gratitude to Dr. Priyanka Singh for her continuous guidance, support, and remarkable patience throughout this project. Her trust in my ability to work independently and explore diverse directions in my research has been instrumental in shaping this work.

I also extend my thanks to Dr. Xue Li for his valuable insights and feedback, which contributed meaningfully to the development of this research.

Abstract

This thesis explores the detection of multimodal disinformation using HAMMER, a transformer-based model designed to identify manipulations in image-text pairs. Using the DGM4 dataset, which simulates real-world disinformation through face and text alterations, the study enhances HAMMER's performance through image restoration, emotion flipping, and background-foreground consistency checks. These methods notably improved detection, particularly in cases where manipulated images and text were semantically aligned. The results highlight the value of multimodal reasoning in combating disinformation and suggest future work in integrating external knowledge sources and expanding to broader media formats.

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Chapter 1. Introduction: Disinformation

"Operation Bodyguard, a World War II disinformation campaign intended to conceal the planned location of the D-Day invasion"[1]. The art of war engulfed in deception is one of the prime examples of the power that disinformation holds. Throughout human history, *intentionally* lying or deceiving has been used as a tactic to profit, whether it be to cover up sensitive matters like the JFK shooting [2], falsifying news before elections [3] or monetary gains [4].

The current media's motive is to increase monetary gain, as almost all channels of information have been monetised, as reflected in [5] "The mainstream media's predilection for sensationalism, need for constant novelty, and emphasis on profits over civic responsibility made them vulnerable to strategic manipulation."

In recent years, Social media has become a fundamental part of modern life, and this statement comes as no surprise: "Social media outperformed television as the major news source for young people of the UK and the USA" [6]. The increased use of social media also means it acts as a great source for the spread of disinformation, "it provides fake news producers with a platform to reach a large audience within seconds. Research indicates that falsehoods spread faster than the truth, reaching 1,500 people six times faster, and social media users are 70% more likely to share or retweet falsehoods than the truth. [7]"The impact of misinformation can be destructive to various aspects of our lives, from public health and politics to climate change and economic issues. [8]" Although this paper[8] uses the term "Misinformation", which highly resonates with the general understanding of the term disinformation.

There are many hot terms thrown around in related studies like: Hoax, Fake News, Rumour, Spam, Troll, Misinformation, etc., and their overlapping use has made the distinction in their meanings unclear [9]. This is why it is important to define "Disinformation" if we want to be clear with our goals and further research on this topic.

[2], Fetzer argues that the definition is closely related to the context in which it is being used, alongside how someone who is *misinformed,* believing the untrue information to be true, could act as a source of disinformation unknowingly. Building on Fetzer's and others' work [1] concludes on the definition "*disinformation is misleading information that has the function of misleading."* This is the definition that we will be using throughout this

research, as it provides a clear distinction from other terms by discriminating through intent.

Most forms of online uploads like news, blogs, social media, etc. are mostly multimodal, i.e incorporate more than one type of media, i.e. Image, video, audio, text. This means that there are more convincing ways of deceiving. Two modalities that confirm each other increase the deceiving effects on the perception of the person consuming the media.

This is the motivation to pursue disinformation detection in multimodal media.

Chapter 2. Literature review:

Although developing on the HAMMER model remains the aim of the project, we delve into related topics like the use of AI, user behaviour, social dynamics, etc., because while not being directly involved, their ablations can help us further our understanding and refine our approach.

Tackling Multimodal Fake News

[10] Addressed the challenge of combating multimodal fake news, which involves false information presented through multiple media types. Combining multimodal information significantly improves the ability to detect fake news compared to relying on a single modality. Other than the obvious textual and visual modalities, they also used network-based information, i.e. user activity on the platform, previous posts, etc., as one of the modalities because the paper argues in favour of this information being used in the prediction of fake news detection systems:

Early approaches to fake news detection utilised traditional ML methods such as “Support Vector Machines (SVM)”, “Random Forests (RF)”, and “Naive Bayes (NB)”. However, the growing volume of data has led to a shift towards DL approaches, including “Convolutional Neural Networks (CNN)”, “Recurrent Neural Networks (RNN)”, “Long Short-Term Memory (LSTM)” networks, and “Generative Adversarial Networks (GANs)”.

Early detection of fake news is crucial due to the rapid spread of rumours, and if they can be classified or flagged before widespread it can stop the number of people the disinformation influences, as each person sharing has an exponential effect on the reach of the disinformation piece. [11] proposed a “hybrid CNN-RNN model” that treats news propagation as a multivariate time series, achieving high accuracy in detecting fake news shortly after its emergence.[12] developed a supervised framework to identify promoted campaigns on Twitter by analysing temporal patterns and network structures, achieving a high AUC score for early detection.

Content-based approaches, which analyse the text and images within articles, have generally performed better than context-based approaches, which focus on propagation structures and user data. [13] Introduced a geometric deep learning approach for fake news detection using a Graph Neural Network, integrating network structure and user features developed a framework that considers user biases alongside news content to predict the likelihood of forwarding a post.

Addressing the limitations of content-based detection in microblogging sites, event-level detection models consider broader contexts, including post-repost structures and comments. [14] Employed "Hierarchical Networks with Social Attention (HSA-BLSTM)" to evaluate the credibility of posts within an event. [15] Proposed a framework leveraging "domain-specific and cross-domain knowledge" to detect fake news across various topics.

Evidence-Based Fact-Checking: This approach involves verifying articles against external sources. Manual fact-checking is conducted by experts or crowdsourced, automatic fact-checking employs ML/DL techniques. For example, "CompareNet" [16] uses an "end-to-end graph neural model" to compare news against a knowledge base for automatic verification.

"Multimodal Fake News Detection (MFND)" frameworks assess the authenticity of news by combining both text and image. The general approach involves extracting features from each modality and combining them to evaluate the credibility.

Textual and Visual Channels:

- **Textual Channel:** Features from text are extracted using various word embeddings and pre-trained models. Techniques such as "Bi-LSTM" and "attention mechanisms" are used to capture "long-term dependencies" (referring to context not in the immediate text) and semantic meanings in the text. Commonly used models include "Word2Vec", "GloVe", and "BERT", with some frameworks enhancing semantic analysis through "hierarchical structures" or "graph-based models."

- **Visual Channel:** Features from images are obtained using pre-trained models like "VGG-19" and "ResNet50". Approaches such as image captioning, forensic analysis, and region-based feature extraction are used to address image manipulation and improve feature representation.

Fusion Techniques:

- **Early Fusion:** This technique combines multimodal features at the data level before classification. Methods like "Principal Component Analysis (PCA)" and "Canonical Correlation Analysis (CCA)" are used

for this purpose. However, early fusion can be complex and often requires rigorous data filtering.

- **Late Fusion:** In this approach, separate classifiers are used for each modality, and fusion occurs at the decision level, offering flexibility, but may lead to a loss of correlation between modalities.
- **Intermediate Fusion:** This technique integrates modalities at hidden layers within the model, improving joint representation learning through multiple layers. Methods such as "neuron-level attention" and "co-attention layers" are used to effectively align features from different modalities.

Additional Features:

- **External Knowledge:** To enhance semantic understanding and verify entities in news articles, "real-world knowledge graphs" are incorporated. Techniques such as "entity linking", and "knowledge distillation" are used to fill the feature set with external information.
- **Event-Level Features:** This approach focuses on the broader context of related posts within an event rather than individual posts. Models like "Event Adversarial Neural Networks (EANN)" and "Multi-modal Topic Memory Networks (MTMN)" capture event-specific features and improve generalisation across different events.

Comprehensive Frameworks Combining Machine Learning and Knowledge Engineering

[17] Integrates machine learning with knowledge engineering. This method addresses the limitations of existing approaches that often focus on isolated elements like user metadata or group dynamics, neglecting the actual content of the information, the opposite of the approaches we have discussed till now.

1. A novel "term weighting scheme" enhances the relevance of specific terms within the context of fake news, improving the accuracy of content-based detection.
2. "Clique-based features" use the structural relationships between users to improve detection, recognising that fake news often spreads through specific network patterns.
3. Additional contextual data from users is used to refine the identification process, providing a more "complete" view of the information ecosystem.

Techniques for Improved Accuracy

Ensemble Learning and Deep Learning

[18]Explored various machine learning models for fake news detection, including traditional techniques like "Naive Bayes", "Support Vector Machines (SVM)", and "Decision Trees", as well as more advanced methods like ensemble learning and deep learning approaches. The research found that "ensemble models", which "combine multiple machine learning algorithms, tend to outperform individual models in terms of accuracy and reliability". Specifically, ensemble methods like "Random Forest and Gradient Boosting" showed better results in identifying fake news across various datasets. The authors also investigated "deep learning models", including "Convolutional Neural Networks (CNNs)" and "Recurrent Neural Networks (RNNs)", which demonstrated high accuracy in detecting complex patterns in text data. However, they noted that the success of these models heavily depends on the quality and quantity of the training data.

Deep Learning for Multicategorical Text Analysis

[19] Proposed a deep learning-based framework for detecting misinformation in multicategorical text contents. The model utilises transformers and uses attention mechanisms to improve the interpretability of the model's predictions. The framework is designed to handle the complexity of multicategorical text data by involving a "committee of classifiers". By using transformer architectures, the model can capture long-range dependencies in text, which is crucial for understanding context in misleading information. The attention mechanisms used in the model allow for better interpretability, as they highlight which parts of the text are "most influential in the model's decision-making process". "The proposed approach outperforms the Weighted BERT Ensemble by nearly 30% in Balanced Accuracy and exhibits a 50% improvement in the F1 metric. It also sustains competitiveness with Batch MTL, with a slight 0.8% increase in the F1 score."

Understanding User Behaviour and Social Dynamics

Impact of User Motivations on Correction Effectiveness

[20] Conducted an experimental study to examine the effectiveness of various rightness on belief in misinformation, particularly about users' motivations for using social media. The research compared web add-on corrections (e.g., fact-check labels) with narrative corrections (e.g., detailed explanations). "Web add-on correction was effective in decreasing belief in misinformation across users, regardless of their motivations for using social media."

1. "For users who primarily use social media for social interaction, narrative correction was more effective in reducing misperceptions."

2. The effectiveness of correction methods varied depending on users' motivations for using social media, suggesting that personalised approaches may be beneficial.

These results suggest that social media platforms should tailor their misinformation correction strategies based on users' motivations and the type of content they engage with.

Psychological and Social Mechanisms of Misinformation Spread

[8] Investigated the factors contributing to the spread of misinformation on social media, focusing on psychological and social mechanisms. Their study combined behavioural analysis with machine learning techniques to examine how misinformation spreads.

1. "Misinformation spreads more rapidly when it aligns with users' existing beliefs", highlighting the role of confirmation bias in the propagation of fake news.

2. Cognitive shortcuts: Individuals often rely on mental heuristics when evaluating information, which can lead to errors in judgment.
3. Illusory truth effect: Repeated exposure to false claims increases the likelihood of believing them, even if they were initially identified as false

The authors suggest that social media platforms need to implement stricter content moderation policies and redesign their algorithms to prioritise truthfulness over engagement. They also emphasise the importance of user education to improve critical thinking skills and reduce the impact of misinformation.

The Role of Artificial Intelligence in the Future of Fake News Detection

[21] Explored the role of artificial intelligence in combating fake news and misinformation on digital platforms.

1. Machine learning algorithms: Traditional methods like Support Vector Machines and Random Forests, have shown effectiveness in classifying text-based content.

2. Deep learning models: "Advanced neural network architectures such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks", which can capture complex patterns in data.

The authors highlight the potential of AI to enhance the accuracy and efficiency of fake news detection systems. They also discuss the challenges in implementing AI-based solutions, such as the need for large, high-quality datasets and the importance of addressing bias in AI models. The paper

concludes that while AI shows great promise in combating fake news, it should be used in conjunction with human expertise and critical thinking to achieve the best results.

Limitations and Future Directions:

Limitations:

1. Single-modality focus: Many studies, such as [22] and [23]Focus primarily on text-based misinformation. This approach fails to address the multimodal nature of fake news, which often combines text, images, and videos.
2. Limited interpretability: While machine learning models have shown promise in detecting fake news, many lack interpretability, making it difficult for users to understand and trust the detection process [24].
3. Narrow scope of user behaviour analysis: Studies like [20] examine user motivations, but there's a lack of comprehensive understanding of how various psychological and social factors interact in the spread and belief of misinformation.
4. Overemphasis on detection: There's a tendency to focus on detecting fake news rather than preventing its spread or mitigating its effect [25].
5. Limited consideration of cultural and linguistic diversity: Many models are developed and tested primarily on English-language content, potentially limiting their effectiveness in diverse global contexts.

As Multimodal approaches can capture complex relationships between different types of content, they improve detection accuracy. The development of comprehensive datasets that represent the diversity of multimodal content on social media is crucial for advancing research in this area.

Research Gap:

The lack of multimodal analysis calls for developing more sophisticated models that can effectively analyse and integrate text, image, video, and network data to detect misinformation comprehensively. According to Bilgin & Wang [23] and [8], future research should focus on the psychological and social mechanisms that drive the creation, spread, and belief in misinformation. There is a need to focus on tracking and analysing the spread of misinformation across multiple social media platforms to develop more effective prevention strategies. Building on Lee's[20] findings, research

should explore how to tailor misinformation correction and prevention strategies based on individual user characteristics and motivations.

What we learn here is that there are multiple factors that directly influence the spread of disinformation to varying extents. Suggestions that come out of the literature review also require further studies like psychological behaviour analysis, propagation patterns or integrating further metadata modalities. But the truth is that for the people most susceptible to being influenced by online disinformation, it is not feasible to assume that they can collect this information. Automating such checks through third-party models could lead to privacy/ethics concerns. Linking behavioural patterns is the utmost difficult in ever-changing trends, of which people interact with social media with technological developments.

Idealistically, you would want a model that is capable of detecting disinformation purely of what bare minimum is available, i.e. just the manipulated piece of media. If this model can perfectly classify what is given to it, it eliminates the need for any additional information like propagation patterns, human studies, etc. Although these studies are still necessary in their respect but for our aim of 'Multimodal Disinformation Detection', this is the purest approach to adopt.

With this philosophy, I worked on the SOTA HAMMER model built for the DGM4 dataset [26].

Chapter 3. Theory: HAMMER and DGM4

Here, I summarise the functionality and necessary details required to know about the HAMMER model and DGM4 dataset [26] to understand the basis of my work on it.

3.1 The DGM4 Dataset

The DGM4 (Detecting and Grounding Multi-Modal Media Manipulation) dataset was introduced to simulate real-world multi-modal disinformation, particularly focusing on deceptive combinations of images and text drawn from credible media sources such as *The Guardian*, *BBC*, *USA TODAY*, and *The Washington Post.* It acts as a benchmark for detecting and grounding manipulations across both visual and textual modalities [26].

Unlike prior datasets that are restricted to single-modality manipulations, such as FaceForensics++ for facial forgery detection [27] or LIAR for textual misinformation [28], DGM4 introduces synthetically manipulated yet realistically grounded samples. It targets human-centric news content, which often contains high-impact information concerning political figures,

celebrities, and social narratives. The dataset comprises 230,000 total samples, divided into 77,426 pristine pairs and 152,574 manipulated ones. The manipulated samples are evenly distributed across four primary categories, two for image and two for text manipulation, along with complex mixed-modality cases [26].

Each manipulated instance in DGM4 falls under one or more of four well-defined classes: Face Swap (FS), Face Attribute (FA), Text Swap (TS), and Text Attribute (TA).

Face Swap (FS)

FS manipulations involve replacing the face of the central figure, mostly in an image, with another identity. This is achieved using deep learning-based methods such as SimSwap [29] or InfoSwap [30]. Bounding boxes are annotated around the swapped face for supervised grounding tasks.

Face Attribute (FA)

FA manipulations retain the subject's identity but modify their facial expression, such as changing a smile to a frown, to invert the emotional context of the image. This was achieved by StyleCLIP [31] and HFGI [32]. This manipulation is subtle yet impactful, as it alters sentiment while avoiding outright detection. As with FS, bounding boxes of manipulated regions are labelled.

Text Swap (TS)

TS involves semantic alterations of text by replacing the original caption with another sentence that contains the same named entity (typically a person's name) but describes a different context. The substitution is based on low cosine similarity in semantic space using SentenceBERT [33], ensuring that the new text diverges in meaning. Ground truth annotations label manipulated tokens.

Text Attribute (TA)

TA manipulations aim to subtly shift the sentiment polarity of the caption, transforming a neutral or positive message into a negative one or vice versa. The manipulation is implemented via a sentiment-aware text generation model, BGST. This form of manipulation is particularly deceptive because it introduces minimal cross-modal inconsistency, making detection substantially more difficult.

The dataset comprises 66,722 FS samples, 56,411 FA samples, 43,546 TS samples, and 18,588 TA samples. Approximately 32,693 samples include mixed manipulations such as FS+TS or FA+TA. Notably, manipulations are subtle; most image changes affect less than 10% of the pixel area, and TA

modifications usually involve altering only 1–3 tokens. To simulate noise in natural media distribution, 50% of the dataset is also augmented with perturbations such as Gaussian blur and/or JPEG compression. This also disturbs the usual signs of image manipulation, like poor editing or pixel colour inconsistencies.

3.2 The HAMMER Model

To address the complexity of the DGM4 dataset, HAMMER, HierArchical Multi-modal Manipulation rEasoning tRansformer—a transformer-based architecture was proposed [26]. It is capable of both classification and grounding tasks across visual and textual modalities. The model is composed of two unimodal encoders: Vision Transformer (ViT) [34] for images and BERT for text [35], and a multi-modal aggregator, all integrated through shallow and deep manipulation reasoning stages.

The architecture of HAMMER is divided into 1)Shallow Manipulation Reasoning: Contrastive learning aligns embeddings between modalities, followed by 2)Deep Manipulation Reasoning: A multi-modal aggregator fuses embeddings through cross-attention for richer semantic correlation and 3)Prediction Heads: Cb: Binary Classifier, Cm: Multi-label Classifier (FS, FA, TS, TA), Dv: Bounding Box Detector, Dt: Token-Level Detector. Each head operates at different semantic depths, enabling both high-level (e.g., real/fake) and fine-grained (e.g., manipulated face region) predictions.

Contrastive Learning

In the shallow reasoning phase, HAMMER leverages Manipulation-Aware Contrastive Learning (MACL) using a variant of the InfoNCE loss [36]. Two encoders extract CLS tokens from image and text modalities, projecting them into a joint embedding space. Original pairs are pulled closer, while manipulated or mismatched pairs are pushed apart, highlighting subtle semantic mismatches.

Grounding Manipulations

HAMMER is equipped to localise manipulation regions.

Image Grounding: By cross-attending image and text features, HAMMER identifies bounding boxes (ybox) corresponding to manipulated facial regions. A Local Patch Attentional Aggregator (LPAA) module is employed to enhance spatial focus.

Text Grounding: Tokens are analysed using a deep cross-modal aggregator that provides token-level binary predictions (ytok). Each token embedding reflects contextual and visual alignment, enabling fine-grained grounding.

Classification Tasks

HAMMER simultaneously performs binary classification to determine if an image-text pair is manipulated or not and multi-label classification to predict specific manipulation types. This multi-task setting allows HAMMER to act as an explainable disinformation detection system that not only flags manipulated content but also explains how and where the manipulation occurs.

Chapter 4. Methodology

In this section, the methodology for detecting and grounding multimodal disinformation using the HAMMER model and my proposed enhancements is described. We follow the overall architecture of HAMMER as outlined above and then integrate additional modules into the pipeline. The specific techniques introduced are image denoising, emotion consistency checks, reverse image search, and background–foreground analysis. *Figure 1* illustrates the overall system pipeline, showing how these components fit together from input to output.

Also, in Figure 1 we can see contrastive learning in action through the GRAD CAM visualisation, where words are queried against the image. The sentimental words in the text are compared against the sentiment on the face in the image to make a contrastive-learning decision.

The base HAMMER (HierArchical Multi-modal Manipulation rEasoning tRansformer) model uses PyTorch, following the architecture proposed by Shao et al. (2023). For the image encoder, a Vision Transformer (ViT-B/16) pre-trained on ImageNet is adopted. This model outputs a sequence of patch embeddings, to which a trainable [IMG_CLS] token is appended to capture global semantic information across the image.

For text processing, a 12-layer BERT-base encoder initialised from uncased English pre-trained weights is used [35]. A special [TEXT_CLS] token is inserted at the beginning of each sentence to serve as a global textual representation. Both encoders are configured to output 768-dimensional features, allowing for straightforward cross-modal comparison through dot-product similarity in a shared embedding space.

During the contrastive alignment stage, the [IMG_CLS] and [TEXT_CLS] embeddings are projected into a common 256-dimensional space using linear transformation layers. These projected embeddings were then normalised to unit length before computing the contrastive loss.

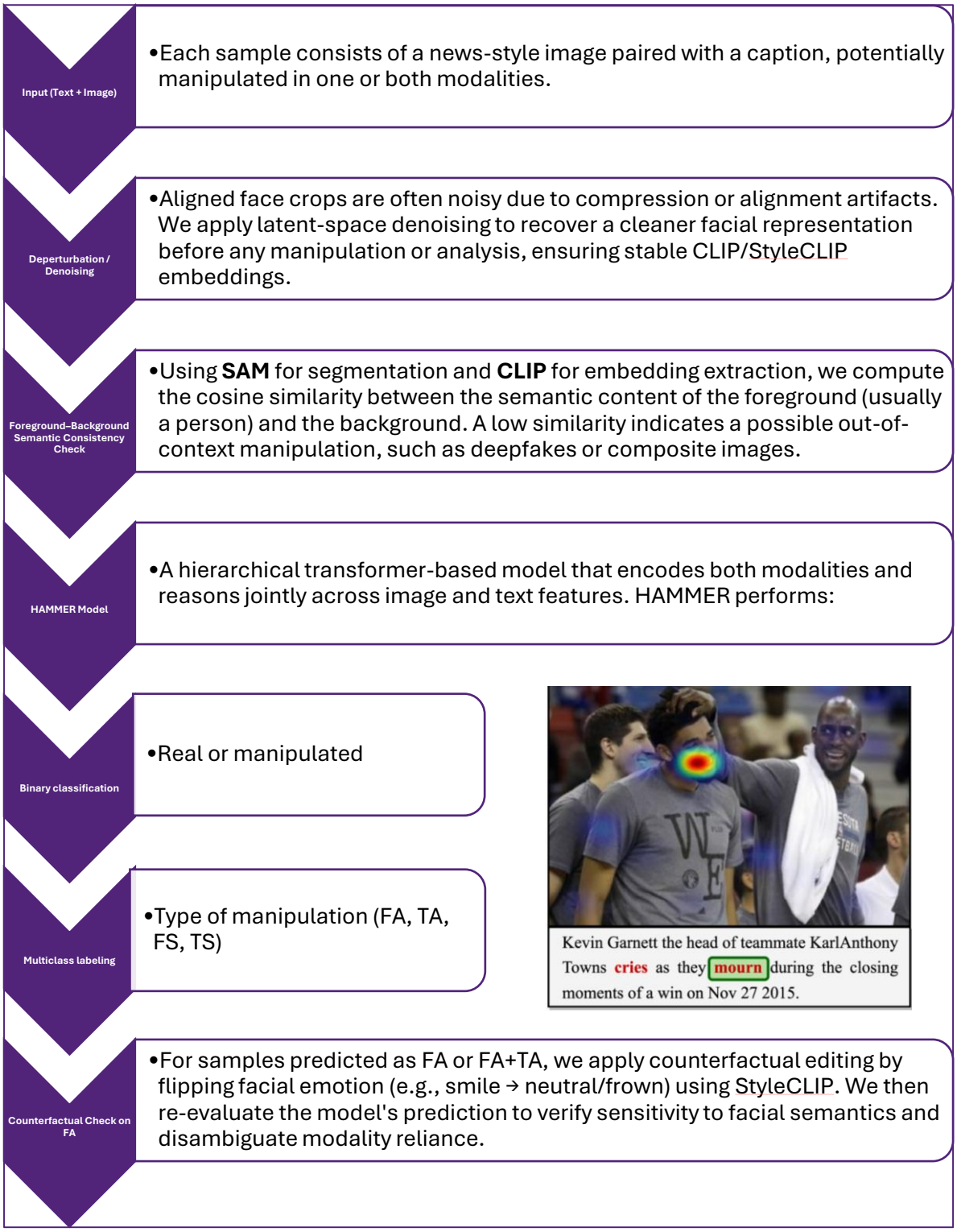

Figure 1Methodology

Visualisation and Output Interpretation

To interpret and present model outputs, I used **Streamlit** to create a visual interface that ingests and displays predictions from the JSON files generated during inference. These files contain outputs for binary classification, multi-

label classification of manipulation types, predicted bounding boxes for manipulated image regions, and identified manipulated tokens in the text.

For interpreting classification confidence, I applied a softmax function over the multi-label prediction logits to extract the probability of each manipulation category.

4.1 Denoising and Deblurring

Samples in the DGM4 dataset often exhibit varying degrees of noise, blur, and compression artifacts, which degrade visual clarity and compromise downstream processing tasks. These distortions not only affect image-text alignment but also hinder tasks like emotion inversion and facial expression manipulation, both of which require fine-grained visual features and were used in section 4.2. Moreover, clear images are important for enhancing reverse image retrieval accuracy and facilitating more reliable grounding between modalities.

To address this, we implemented a two-stage restoration pipeline. First, we identified the nature of the image distortion, categorising it as either Gaussian blur, motion blur, or general noise, using basic edge loss metrics and visual inspection. Then, we applied Real-ESRGAN [37], a generative adversarial network-based super-resolution model, to restore noisy samples. This was followed by Restormer [38], a Transformer-based image restoration model specifically trained for motion deblurring, denoising, and deraining tasks. The combination of Real-ESRGAN and Restormer, improve image fidelity, allowing better input conditions for both manipulation detection and facial emotion flipping.

By preprocessing all image samples in this manner before feeding them into the HAMMER model, I aimed to reduce false detections caused by compression noise and restore images for manipulation localisation.

4.2 Emotion Flipping

One of the trickiest forms of multimodal disinformation is when both image and text are manipulated in a mutually reinforcing manner. This alignment creates high semantic coherence between misleading modalities, thereby maximising the credibility and persuasive potential of the disinformation. As highlighted in the HAMMER paper itself, such aligned manipulations, particularly involving facial expressions are exceptionally challenging to detect, even for robust multi-modal models. This is where the HAMMER model is the weakest in, identifying FA+TA.

Such alignment interferes with HAMMER's contrastive learning objectives by minimising cross-modal feature discrepancy, making it harder for the model to flag inconsistencies.

To explore this, I performed counterfactual emotion flipping on FA and FA+TA samples. Specifically, intervening with the detected facial emotion by flipping to its opposite emotion in the image while keeping the text unchanged. This forced the image-text pair into a mismatched or misaligned configuration, which, when passed through HAMMER again, allowed us to assess the model's sensitivity to emotional grounding.

We employed the STYLECLIP [31] framework, which uses latent manipulation in the StyleGAN [39] latent space guided by CLIP-based directional vectors, to perform facial expression inversion. The face regions were first isolated and inverted using the pipeline proposed in the DGM4 construction, which was further enhanced by the integration of the pre-crop emotion denoising in section 4.1.

Effectively, once the HAMMER model detects something as FA manipulation we preserve the text from the input, flip the emotion on the face in the image. This generates a new input pair for the HAMMER model with the original text and now edited image. This means this time if the model missed text manipulation due to semantic consistency, it is forced to recognise it this time leading to increased accuracy and detection.

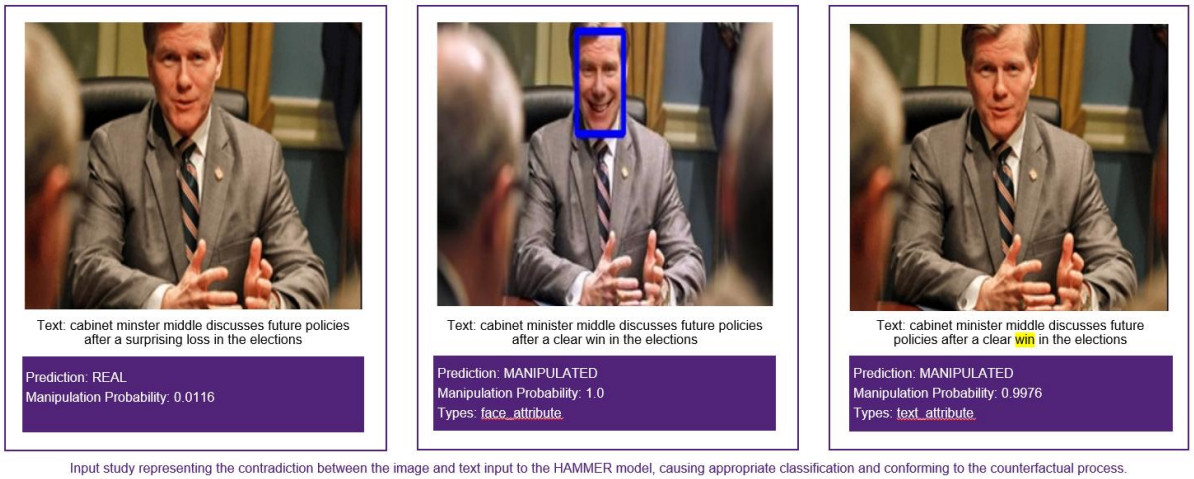

Figure 2 Example of emotion flipping

The original HAMMER model achieved 81.25% accuracy on FA+TA (face + text attribute) manipulated samples from the DGM4 dataset. After applying counterfactual image flipping (altering facial expressions to contrast or align with accompanying text), accuracy increased to 83.522%, yielding a 2.272% absolute improvement.

4.3 Reverse Image Search

To further assess binary classification of manipulated images, I studied images using Google's reverse image search. This allows to locate potential original or undisturbed versions of manipulated images from open web sources, thereby providing a qualitative baseline for comparison.

Using Google reverse image search to query 100 images in the DGM4 dataset and recorded the top 1–3 results, prioritising exact matches. The retrieved original images served two key functions.

The fact that the DGM dataset is based upon news samples it is not at all difficult to reverse google image search to find ground truth and this is not at all the case if the images are part of social media uploads or uploaded by individuals on lesser-known platforms. The perturbations make it somewhat challenging to perfectly match images even after denoising such as in this image.

Figure 3FA example from DGM4

Figure 3 is an FA example with not much clues to pick up on. The face in focus is swapped itself and the image itself is perturbed. Barring such examples.

Whereas examples where model fails, results show up in exact matches. Figure 4 shows that the HAMMER model classifies this image as FA only but upon manual inspection, shown in figure 5, we see it is a case of both FA+TA.

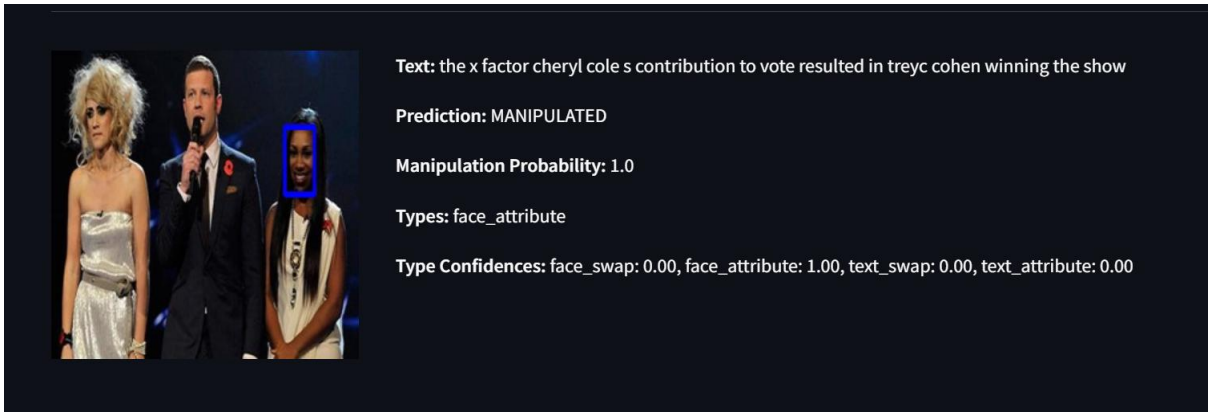

Figure 4 False classification of FA+TA

The X Factor: Cheryl Cole's refusal to vote resulted in Treyc Cohen (right) leaving the show. Photograph: Ken McKay/Rex Features/Talkback Thames/ITV

Figure 5 Original image

4.4 Background Foreground Check

While the DGM4 dataset does not explicitly contain manipulations involving background replacement, I implemented an experimental pipeline to test whether the model could detect inconsistencies between a foreground object (typically a person) and its surrounding background. This is particularly

important for extending HAMMER's capabilities to future datasets were compositional manipulations, such as pasting a subject into a misleading scene may be more prevalent.

I used the Segment Anything Model (SAM) [40] to extract a high-quality foreground mask from each image. The SAM model, trained on a diverse set of objects and scenes, reliably segmented human subjects and primary objects from their respective scenes. Once the foreground and background regions were isolated, I extracted CLIP embeddings for both components independently.

I then computed the cosine similarity between the foreground and background embeddings to assess semantic coherence. A low similarity score indicates potential inconsistency or compositional manipulation.

One hypothetical application of this technique would involve detecting manipulated backgrounds in political imagery—for instance, an image of a political leader like Donald Trump placed against a backdrop of a Volcano. Even if no pixel-level edits are apparent, the semantic mismatch can provide strong cues for disinformation. The improvement to HAMMER model does not include foreground-background (FG-BG) semantic inconsistencies, which are not defined as manipulation types in DGM4. DGM4, built from news-style samples, limits manipulation classes to face/text attribute or swap edits. Therefore, FG-BG consistency checks are not reflected in standard accuracy metrics but serve as a complementary signal for identifying out-of-context visual manipulations that HAMMER currently misses.

Figure 6 HAMMER model for trump-volcano

HAMMER model fails to classify example in figure 6, but as from figure 7 such inputs can be detected as a pre-processing step.

Figure 7 SAM model separation visual

SAM model separates the foreground and the background of the image. The CLIP model is then used to extract embeddings. Then we calculate the cosine similarity between the two. In this case we get a low cosine value of 0.31.

What we infer from this is "(Donald Trump)a person wearing a suiting probably a politician foregrounds a bursting volcano does not seem right"

Chapter 5: General Discussion and Conclusion

Here I write down some interesting observations about the HAMMER model and this architecture.

Generalization to Unseen Domains: HAMMER model is trained on a specific dataset (DGM 4) which focuses on human-centric news. If we try to apply it to a different domain (say medical images with captions), it does not generalize well. HAMMER model still overfits to the style and content of DGM4 news. For example, in DGM4 most images are photographs; if given a cartoon or infographic with text, the model behaves unpredictably. Additionally, if an image is manipulated in a way not seen in training (e.g., a new type of DeepFake technique), the model might miss it or misclassify it. We did observe that the model struggled with extremely subtle manipulations that were *technically* there but almost indiscernible (like a single word synonym change that doesn't alter meaning much, or a facial alteration that's within natural variation). This raises the question: should those even be flagged? Generalizing what counts as disinformation is tricky.

Data Bias and Limitations: The dataset itself has biases. For instance, all the manipulations in the DGM4 dataset are synthetic and some of it obviously mismatched (TS), which the model learns to catch. In the real world, fake news is more subtly written by humans to sound convincing, HAMMER model

might find those harder if they don't violate image consistency clearly as it already struggles the most in FA+TA class. Also, DGM4's "real" examples are genuine news – but in reality, even genuine news can have varying styles and not always perfect image-caption alignment. The model might misfire on a real news piece that happens to have an unusual image-caption relationship (not due to deceit, but just editorial choices). For example, a headline might be metaphorical while the image is literal, not a fake, but the model could flag inconsistency. This relates to the model's lack of world knowledge: it doesn't truly understand context or idioms.

Lack of World Knowledge and External Fact-Checking: Our model is constrained to the content of the image and text. It has no access to databases of facts. If both the image and text present a coherent story that is entirely fabricated, the model sometimes classifies it as "real" because it finds no inconsistency. This is a fundamental limitation: consistency-based detection cannot catch fake news that is self-consistent but factually false. Addressing this requires integrating **verified knowledge** or performing cross-source verification (e.g., checking if the event was reported elsewhere, if the person was actually present, etc.). This work scratched the surface of this by reverse image search for known context, but that's not a full solution. Future models might need to interface with knowledge graphs or the web to validate claims, going beyond just the image-text pair in isolation.

Computational Complexity: The HAMMER model with transformers and multiple heads is heavy inference is not real-time for high-resolution images. Our additional modules (especially image enhancement and segmentation) add to the computational load. In a deployment scenario, this could be a hurdle if scanning large volumes of content. There is a trade-off between accuracy and speed. It can be mitigated to some extent by using smaller versions of tools like the "anime" Real-ESRGAN model for speed at slight cost of quality, etc... Nonetheless, the pipeline is not light. Model compression or distillation could be necessary to use this in, say, a browser plugin or mobile app.

Multi-Modality: This thesis focused on static images and text. But fake news can involve videos, audio, etc. Extending to video and text brings challenges of temporal consistency (did the video depict what the caption claims throughout?). This architectural approach could theoretically extend (treat frames as image sequence, etc. Also, some news may include multiple images or a gallery with a text article, this model handles one image and caption at a time. Scaling to many-to-many (multiple images with a body text) is an open challenge. Possibly it could analyse each pair and then aggregate the results.

In summary, current models like HAMMER are powerful but not foolproof. They rely on learned correlations and can miss cases if the cues are subtle or if the manipulation lies outside their training distribution. They can also be overly sensitive to the presence of certain artifacts, for instance, if a manipulation method leaves a signature (like a slight blurriness), the model might latch onto that, A real image with coincidental blur could be a false positive.

A practical hurdle in deployment is that adversaries can adapt. If they know models like this check for image-text consistency, they might try to manipulate both modalities consistently. Or if they know we do denoising, they might introduce counter-forensic measures (like noise patterns that fool the model after denoising).

My earlier hypothesis of one model to solve all issues without the need of additional queues stands corrected. With current SOTA models we still need external verification for absolute classification confidence.

Conclusion

In this thesis, we explored the detection and grounding of multimodal disinformation using the HAMMER model on the DGM4 benchmark. This work confirms that multimodal approaches are not only necessary but highly effective for catching false narratives that span text and image. By analysing both modalities jointly, the model can uncover inconsistencies that neither modality alone could detect – a significant advancement over prior single-modal fake news detectors. We demonstrated that the model can accurately classify manipulated image-text pairs and, in many cases, provide human-interpretable explanations by highlighting the fake components in the image and text.

To the extent of contribution,1) proposed an emotion flipping method to highlight cases where the text and image align, 2) proposed integration of a basic check, foreground background coherence, and, 3) simple pre-processing to aid inversions of images to assist further tasks.

Future Work

Building on this thesis, there are several avenues to pursue:

- **Integrating Fact-Checking Knowledge:** Combining multimodal detection with knowledge graphs or web retrieval to verify factual claims could address the issue of coherent-but-fake stories. For instance, if the text says, “President X visited Y”, the system could query a knowledge source to see if that event actually happened.

- **Real-time and Large-Scale Deployment:** Optimizing the model for speed (perhaps via model compression or by simplifying the architecture) and testing it on live social media data would be valuable. This includes handling streams of posts and possibly filtering priorities (e.g., flagging only high-impact suspicious content to minimize noise).
- **User Interface and Explainability:** How to present the model's output to end-users or fact-checkers is key. Developing intuitive visualizations (like highlighting image regions on a news article interface or providing an "inconsistency report") could make the tool more practical. Also, incorporating user feedback (if a user indicates an error, the system could learn from it) might improve reliability.
- **Robustness to Adversarial Manipulations:** As detection models improve, adversaries might intentionally try to fool them (adversarial examples or designing fakes that exploit model blind spots). Research into making the model robust to such attacks (perhaps via adversarial training, or by constantly updating the model against new fakes) will be important for staying ahead in the arms race against disinformation.
- **Broader Modalities:** Extending these techniques to video+text (multimodal misinformation in video captions or video news) and audio (audio transcripts vs video) is a natural next step. Some recent multimodal Transformers can handle text, image, and even audio adapting those for fake detection would broaden the coverage.
- **Ethical and Policy Considerations:** Finally, any deployment of such technology should consider the ethical implications. False positives can damage credibility or censor real content, and false negatives let fakes slip through. There's also the risk of deepfake "detection" tools being used maliciously (e.g., to falsely claim real content is fake). Ongoing research in watermarking AI-generated media and in calibrated model confidence will complement detection efforts to create a more trustworthy information ecosystem.

Appendix A: Program listings

The following configuration was used during testing: image_res = 256, vision_width = 768, embed_dim = 256, batch_size_train = 64, batch_size_val = 256, temp = 0.07, queue_size = 65536, momentum = 0.995, alpha = 0.4, max_words = 50, label_smoothing = 0.0.

The following loss weights were used: loss_MAC_wgt = 0.1, loss_BIC_wgt = 1, loss_bbox_wgt = 0.1, loss_giou_wgt = 0.1, loss_TMG_wgt = 1, loss_MLC_wgt = 1, optimizer = adamW, lr = 1e-4, weight_decay = 0.02, schedular = cosine, epochs = 50, min_lr = 1e-6, decay_rate = 1, warmup_lr = 1e-6, warmup_epochs = 10, cooldown_epochs = 0

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